

Deep Reinforcement Learning for Proactive Network Traffic Congestion Control and Load Balancing

Course Proposal — Deep Learning & Neural Networks

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June 16, 2026

1. Problem Statement

Modern telecommunications and enterprise networks suffer from recurring **traffic congestion** — a condition where aggregate demand for bandwidth exceeds available capacity. When congestion occurs, packet loss rises, latency spikes, and end-user quality of experience (QoE) degrades sharply. Current mitigation relies heavily on human network optimizers who *reactively* redistribute load across links and nodes. This manual, rule-based approach is slow to respond, fails to generalise across heterogeneous topologies, and cannot anticipate traffic bursts before they manifest as outages.

2. Proposed Solution

We propose a **Deep Reinforcement Learning (DRL)** agent — specifically a Proximal Policy Optimisation (PPO) or Deep Q-Network (DQN) model — trained to perform *proactive* traffic load balancing. The agent will observe real-time network telemetry (link utilisation, queue depths, flow statistics) as its state, and output routing or traffic-shaping actions that redistribute load *before* congestion onset. Unlike reactive heuristics, the DRL policy learns a generalised balancing strategy from simulated traffic episodes, enabling autonomous, millisecond-scale decisions without operator intervention.

3. Objectives

1. Build a network environment simulator (NS-3 or graph-based OpenAI Gym) modelling realistic traffic burst scenarios.
2. Formulate congestion control as a Markov Decision Process (MDP) with a reward function that jointly maximises throughput and minimises latency.
3. Train and compare DRL agents (DQN, PPO) against an Equal-Cost Multi-Path (ECMP) routing baseline.
4. Evaluate convergence behaviour, sample efficiency, and generalisation to unseen network topologies.

4. Methodology

The project follows three phases. **Phase 1** constructs a simulated network topology with configurable traffic matrices drawn from real-world distributions (CAIDA traces). **Phase 2** designs the DRL agent: the state vector encodes per-link utilisation and flow-level features; the action space represents discrete routing weight adjustments; reward is a weighted sum of inverse packet-loss rate and inverse mean end-to-end delay. **Phase 3** conducts comparative experiments measuring average link utilisation, 95th-percentile latency, and congestion incidence rate across multiple topology seeds.

5. Expected Outcomes

The DRL agent is expected to achieve lower congestion frequency and improved QoE metrics relative to the ECMP baseline, demonstrating that learned policies can generalise load-balancing decisions across dynamic, unseen traffic conditions. Deliverables include a trained model, reproducible simulation code, and a technical report summarising findings.

6. Relevance to Course

The project directly applies core course concepts: neural function approximation, temporal credit assignment via the Bellman equation, policy gradient methods, and experience replay buffers. It bridges theoretical deep learning with a practical, high-impact systems engineering problem in telecommunications.